

QUANTS AND MARKET EFFICIENCY

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ABSTRACT. We study the impact of quantitative trading strategies (quants) on price efficiency by developing a model that incorporates two defining features of quants: superior information processing capacity and limited strategic awareness. These forces have opposing effects on trading aggressiveness—better information processing increases it, while limited strategic awareness decreases it. As a result, an increase in quant participation deteriorates price efficiency when their information quality is low, improves it when information quality is high, and produces a non-monotonic effect in between.

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1. Introduction

Advances in artificial intelligence (AI), along with the availability of big data and computing power, have changed how investment decisions are made. These developments have enabled a new generation of investors, commonly known as *quants*, who use quantitative models and algorithms to identify trading opportunities, construct portfolios, and execute trades at scale. While quantitative investing was once limited to a niche subset of hedge funds, it has become increasingly mainstream over the past two decades. Today, it is widely accessible, even to retail investors, through quantitative trading platforms and mutual funds (e.g., Beggs and Hill-Kleespie 2025).

As quants take up a larger share of trading activity, concerns have emerged about how their collective behavior might affect market stability.¹ Episodes of extreme price dislocations such as the August 2007 Quant Meltdown and the May 2010 Flash Crash have raised questions about whether these strategies can, at times, contribute to market stress. While they are often praised for their discipline and ability to eliminate behavioral biases, an emerging body of evidence suggests that quants trading on similar signals and models may react simultaneously to the same events, resulting in crowded trades, fire sales, and feedback loops that can destabilize markets (e.g., Khandani and Lo 2011; Cai et al. 2019; Falato et al. 2021).

The growing presence of quants raises important questions about how they influence other investors and the functioning of markets. Unlike traditional discretionary investors who rely on human judgment, quants tend to process information differently and trade based on systematic strategies. As more quants enter the market, the strategic environment in which prices are formed changes. Yet we know little about how differences in information processing and strategic awareness between quants and discretionary investors influence markets. This paper addresses two key questions: What strategic forces influence investor behavior and price formation as quants and discretionary investors compete? And how does the growing participation of quants impact market efficiency?

¹ According to the Tabb Group, quantitative hedge funds accounted for 31% of all U.S. stock trades in 2019. See, "The Making of the World's Greatest Investor," The Wall Street Journal, November 2, 2019.

To address these questions, we develop a theoretical model in the tradition of Kyle (1985) that incorporates both quants and discretionary investors. The model captures key differences between these two trader types. While quants possess superior information processing capabilities, they lack the strategic sophistication of discretionary investors (e.g., Abis 2022; Kirilenko and Lo 2013). This reflects a common feature of many algorithmic strategies, which are often built to exploit signal-driven patterns without accounting for second-order strategic feedback.

In the model, there are two investment firms (firm A and B), noise traders, and competitive market makers. In the baseline model, we assume firm A is fully discretionary, and firm B consists of a quant and a discretionary department. Before trading begins, nature determines whether the quant or discretionary department of firm B is active. The probability that the quant department is active captures the quantitative investing level in the economy.

The quant department has superior information processing capacity: it observes both the public signal v_1 and a noisy signal θ_2 about a second component of the asset's value, $v = v_1 + v_2$. In contrast, the discretionary department observes only v_1 . Despite the informational advantage, the quant is strategically weaker. It optimises trades based on signals but incorrectly perceives that firm A does not adjust its strategy in response, i.e., it is strategically myopic. In the baseline model, we assume complete strategic myopia, but later relax this assumption to allow for partial responsiveness.

We show that the strategic interactions between quants and discretionary investors give rise to three key effects. First, as the quantitative investing level increases, firm B becomes more effective at extracting information from the additional signal θ_2 and increases its trading aggressiveness on it, which we refer to as the *processing capacity effect*. Second, firm B reduces its aggressiveness on v_1 due to its inability to account for how its trades influence firm A 's behavior—this reflects the *strategic myopia* of quants. Third, the fully discretionary firm A responds by trading more aggressively on the public signal v_1 to exploit firm B 's strategic weakness, which we term the *internalising effect*.

The net effect on overall trading aggressiveness depends on the relative strength of these effects and varies with quants' additional signal quality (q) and quantitative investing level (γ). We show that the processing capacity effect increases linearly with q . Furthermore, the strategic myopia and internalising effects are both independent of q but vary with γ . As γ increases, the myopia effect decays faster than the internalising effect, implying the greatest reduction in trading aggressiveness at low levels of γ .

These mechanisms have non-monotonic effects on overall trading aggressiveness of traders. When q is low, the strategic myopia effect dominates at all levels of γ , leading to a decrease in overall aggressiveness. When q is high, the positive effects dominate and trading aggressiveness increases with γ . At intermediate values of q , the relationship is U-shaped: overall aggressiveness first decreases, and then increases. This result shows that the impact of growing quant participation in markets hinges on both their informational advantage and their ability to account for strategic feedback.

Since prices are determined by order flow in our model, changes in trading behavior naturally affect price efficiency. Following Kyle (1985) and related work (e.g., Subrahmanyam 1991; Holden and Subrahmanyam 1992), we define price efficiency as the inverse of the conditional variance of the asset's value given the price. We show that, similar to trading aggressiveness, when q is low, increasing γ always harms price efficiency. When q is high, it always improves it. At moderate q , the relationship is again U-shaped: price efficiency first decreases, and then increases once the quantitative investing level γ exceeds a threshold. This U-shaped pattern implies that whether quants improve or harm price efficiency depends not just on how much information they process, but also on how well they account for strategic feedback.

We extend the model in two directions. First, we consider a model where both firms have a quant and a discretionary department. This introduces a fourth effect, which we term the *competition effect*. As the quantitative investing level of firm B increases, it competes more aggressively on the additional signal θ_2 , leading firm A 's quant department to reduce its aggressiveness on θ_2 . Therefore, the impact of increasing quantitative investing of one firm depends critically on the composition of the other firm.

In the special case when firm A is already fully quant, increasing the quantitative investing level of firm B still improves its ability to extract information from θ_2 , but this comes with more strategic myopia and more competition for the signal. Moreover, because firm A no longer has a discretionary department, it cannot respond strategically to exploit firm B 's myopia, weakening the internalising effect. Under these conditions, we show that an increase in firm B 's quantitative investing level harms price efficiency, even when the signal quality is high. This suggests that the more quant-driven the rest of the market is, the more likely it is that further increases in quantitative investing by an individual firm harm price efficiency.

Second, we relax the assumption of complete strategic myopia in the baseline model by allowing firm B 's quant department to be partially myopic, parameterised by $h \in (0, 1]$ which represents the degree of strategic myopia; $h = 1$ corresponds to the baseline (fully myopic) model and $h = 0$ corresponds to a fully strategic quant. As h decreases, both the strategic myopia and internalising effects weaken and vanish as h approaches zero, but the processing capacity effect is independent of h . Consequently, the impact of increasing the quantitative investing level γ on trading aggressiveness and price efficiency increases when h is low or q is high, and decreases when h is high or q is low. Therefore, the direction of the effect is driven by the ratio q/h .

1.1. Related literature. Our paper contributes to the emerging literature on the interaction between machines and humans in financial markets (e.g., Abis 2022; Malikov and Pasquariello 2025; Beggs and Hill-Kleespie 2025; Barbopoulos et al. 2024; Coleman, Merkley and Pacelli 2022; Cao et al. 2024). We develop a model that isolates four effects arising from the strategic interaction between machine-based quants and human discretionary investors: the *processing capacity effect*, driven by quants' superior ability to extract information; the *strategic myopia effect*, reflecting their limited awareness of how their trades influence others; the *internalising effect*, capturing the strategic response of discretionary investors to quants; and the *competition effect*, which emerges when multiple quants compete for the same signals.

Our analysis builds on and complements recent work by Abis (2022) and Malikov and Pasquariello (2025). Abis (2022) documents that quantitative and discretionary mutual funds differ in their investment styles. Quants focus on stock picking and process more information, while discretionaries have superior macro-timing ability and portfolio flexibility. This reflects a trade-off between processing capacity and strategic flexibility. Our model formalizes this trade-off in a strategic setting by showing how quants' information advantage can be offset by their inability to anticipate and respond to other investors' behavior.

Malikov and Pasquariello (2025) also develop a model of strategic trading, where quants differ from discretionaries by sticking to backtested strategies. They show that quants can help or hurt market quality, depending on how aggressively they trade. Our model takes a similar approach but differs in two key ways. First, we model strategic myopia not as adherence to a fixed rule, but as an incorrect belief about the opponent's strategy, which allows quants to choose profit-maximising responses given their misperceptions. Second, we introduce a richer setting with two signals and random assignment between quant and discretionary traders, which lets us isolate how different effects impact market outcomes.

Empirically, Beggs and Hill-Kleespie (2025) find that fire sales by quantitative mutual funds have a far greater impact on market instability than those by discretionary funds, largely due to crowded positions and similar trading behavior. Our model helps rationalize this result by showing how strategic myopia leads quants to trade aggressively on signals without accounting for the broader strategic effects, which invites exploitation by discretionary investors and amplify price distortions under stress.

More broadly, our paper contributes to the literature on the effects of automation in financial markets (e.g., Hendershott, Jones and Menkveld 2011; Weller 2018; Dugast and Foucault 2018; Farboodi et al. 2022). We add to this literature by theoretically investigating the impact of quantitative investing, a relatively underexplored form of algorithmic trading, on price efficiency.

The rest of the paper is structured as follows. Section 2 presents and solves the baseline model. Section 3 extends the model to two quant firms. Section 4 concludes. Appendix A contains the proofs, and Appendix B extends the model to allow for partial strategic myopia.

2. Model

We consider a Kyle-type model in which quantitative and discretionary traders coexist. The model allows us to study their interaction and examine how the increase in quant participation affects price efficiency.

2.1. Setup. Three groups of participants trade a risky security: (i) market makers, (ii) noise traders, and (iii) two investment firms (firms A and B). The liquidation value v of the security is given by $v = v_1 + v_2$, where v_1 and v_2 are independent and identically distributed normal variables with mean zero and variance Σ_0 . Each investment firm consists of a discretionary department D and a quant department Q . The discretionary department observes v_1 and trades akin to the rational Kyle (1985) insider. The quant department additionally observes a noisy signal $\theta_2 = v_2 + \epsilon$ with $\epsilon \sim N(0, \sigma_\epsilon^2)$ about the second component of the liquidation value.

We first investigate the case in which firm A is fully discretionary, while firm B has a discretionary department D and a quant department Q . We solve the general model, where both firms have both departments, in Section 3. The model has three dates, $t \in \{0, 1, 2\}$, as shown in Figure 1. At $t = 0$, nature determines which department of firm B participates in the market. The quant department is active with probability γ , and the discretionary department with probability $1 - \gamma$, where $0 \leq \gamma \leq 1$ measures the level of quantitative investing. The realisation of department Q 's or department D 's presence is only observed by firm B .

[Figure 1 here]

Trading occurs at $t = 1$. Let p and w denote the price and aggregate order flow at $t = 1$, respectively. We begin by assuming that traders rationally conjecture the price function as $p = \lambda \cdot w$. Given this conjecture and the opponent's demand x_j , each trader $i \in \{A, D, Q\}$

endowed with an information set ϕ_i submits a market order of size x_i to maximize expected trading profits,

$$E[(v - p(\phi_i, x_i, x_j))x_i | \phi_i, x_i]. \quad (1)$$

For the discretionary departments, the information set is $\phi_A = \phi_D = v_1$. For the quant department, it is $\phi_Q = \{v_1, \theta_2\}$. The additional signal θ_2 captures the greater information processing capacity of the quant department. The quality of the additional signal is $q = \Sigma_0 / (\Sigma_0 + \sigma_\epsilon^2)$.

Although the quant department processes more information, it is strategically weaker than the discretionary department. To capture this, we assume that the quant department of firm B ignores the fact that its trading can reduce the aggressiveness of firm A . While the actual demand of firm A is

$$x_A = \frac{E(v | \phi_A)}{2\lambda} - \frac{E(x_B | \phi_A)}{2}, \quad (2)$$

the quant department incorrectly perceives it as

$$x_A = \frac{E(v | \phi_A)}{2\lambda}. \quad (3)$$

This reflects the strategic weakness of the quant department, as it ignores its impact on the opponent's behavior.

Noise traders trade according to their liquidity or hedging needs exogenous to the model. We assume they demand $u \sim N(0, \Sigma_0)$ units for simplicity. Market makers observe the aggregate order flow (sum of all traders' demands) as

$$\omega = x_A + \gamma \cdot x_Q + (1 - \gamma) \cdot x_D + u. \quad (4)$$

They are risk neutral and competitive. Therefore, the equilibrium pricing rule satisfies weak market efficiency and allows them to break even in expectation, i.e., $p = E[v | \omega]$.

2.2. Discussion of assumptions. Most of the assumptions in our setup are standard and follow the tradition of Kyle (1985). The model introduces two non-standard assumptions to capture key differences between quants and discretionary investors.

First, we assume that quants have access to an additional noisy signal θ_2 about a second component of the asset's value. This reflects their superior ability to process richer or higher-dimensional data sources. Industry reports document the growing reliance of quantitative funds on non-traditional data sources (e.g., news feeds, social media, communications metadata, satellite imagery, and geo-spatial data) to gain an informational advantage.² A similar structure appears in Easley, O'Hara and Yang (2013), who model hedge funds as having access to additional information and unique investment opportunities compared to traditional investors.

Second, we assume that quants are strategically myopic. They optimise trades based on their additional signals but fail to internalise how their actions affect discretionary investors. This captures the idea that many quantitative strategies are designed to exploit data patterns without accounting for second-order strategic feedback. Abis (2022) also argues that humans are more flexible in adapting to changing market conditions, whereas traditional rule-based programming limits the adaptability of quantitative strategies. While we push strategic myopia to the extreme in the baseline model for clarity, we show in the extended model in Appendix B that as long as quants remain less strategic than discretionary investors, our key results continue to hold.

2.3. Equilibrium. The equilibrium concept is a perfect Bayesian Nash equilibrium, defined as follows:

- (i) Each trader i chooses a trading strategy $x_i(\phi_i, E(x_j))$ that maximises expected trading profit, given their information set ϕ_i and beliefs about the opponent's demand $E(x_j)$;
- (ii) The market maker chooses a pricing strategy $p(\omega)$ that breaks even in expectation;
- (iii) Traders and the market maker use Bayes' theorem to form beliefs.

To derive the equilibrium, we start with the optimisation problem defined in Eq. (1) and derive the first-order conditions. Solving the optimisation problem of firm A yields its best response as $x_A = \alpha_A v_1$, where

² See, for example, [Deloitte's InFocus report on 'Alternative data adoption in investing and finance'](#).

$$\alpha_A = \frac{1}{2\lambda} - \frac{\gamma\alpha_Q}{2} - \frac{(1-\gamma)\alpha_D}{2}. \quad (5)$$

For the quant department of firm B , the optimal strategy is $x_Q = \alpha_Q v_1 + \beta_Q \theta_2$, where

$$\alpha_Q = \frac{1}{4\lambda}, \quad \beta_Q = \frac{q}{2\lambda}. \quad (6)$$

For the discretionary department of firm B , the optimal strategy is $x_D = \alpha_D v_1$, where

$$\alpha_D = \frac{1}{2\lambda} - \frac{\alpha_A}{2}. \quad (7)$$

Combining Eqs. (5)-(7) with $\lambda = \text{cov}(v, \omega) / \text{var}(\omega)$ yields a system of five equations and five unknowns $(\lambda, \alpha_A, \alpha_Q, \beta_Q, \alpha_D)$. The following proposition presents the solution to this linear system and summarises the equilibrium outcomes in the presence of both discretionary and quantitative firms.³

Proposition 1. *There is a unique linear equilibrium if an investment firm is fully discretionary, and the other firm has a quantitative investing intensity of γ . The equilibrium pricing rule of market makers is characterised by $p = \lambda\omega$, and the equilibrium trading strategies of traders A , Q , and D are given by $x_A = \alpha_A v_1$, $x_Q = \alpha_Q v_1 + \beta_Q \theta_2$, and $x_D = \alpha_D v_1$, where*

$$\lambda = \sqrt{\frac{(\gamma+4)(\gamma+2)}{4(\gamma+3)^2} + \frac{q\gamma(2-\gamma)}{4}}, \quad (8)$$

$$\alpha_Q = \frac{1}{4\lambda}, \quad \alpha_D = \frac{\gamma+4}{4\lambda(\gamma+3)}, \quad (9)$$

$$\beta_Q = \frac{q}{2\lambda}, \quad \alpha_A = \frac{\gamma+2}{2\lambda(\gamma+4)}. \quad (10)$$

2.4. Quantitative investing and price efficiency. We now examine how the growth of quantitative investing affects the trading aggressiveness of investment firms and the informational efficiency of prices. To be able to compare, we normalise each firm's trading aggressiveness by dividing by the benchmark level $1/3\lambda$, which corresponds to the trading aggressiveness when both firms are fully discretionary. This leads to

³ The linear equilibrium in the model exists uniquely because we adopt the Kyle (1985) framework, and particularly because we assume normality of the exogenous random variables.

$$\tilde{\alpha}_A = \frac{3\gamma + 6}{2\gamma + 6}, \quad \tilde{\alpha}_B = \frac{3}{\gamma + 3}, \quad \tilde{\beta}_B = \frac{3q\gamma}{2}, \quad (11)$$

which represents the normalised trading aggressiveness of firm A on signal v_1 , and of firm B on signals v_1 and θ_2 , respectively. We define overall trading aggressiveness as

$$\tau \equiv \tilde{\alpha}_A + \tilde{\alpha}_B + \tilde{\beta}_B \quad (12)$$

The following proposition shows the effects of quantitative investing on overall trading aggressiveness.

Proposition 2. *When quantitative investing level γ increases, the impact on overall trading aggressiveness τ depends on three effects:*

- (i) *Processing capacity effect (positive): As firm B relies more on its quant department, it trades more aggressively on the additional signal (θ_2) due to improved information processing capacity:*

$$\frac{\partial \tilde{\beta}_B}{\partial \gamma} = \frac{3q}{2}. \quad (13)$$

- (ii) *Strategic myopia effect (negative): The quant department of firm B is less flexible and ignores its strategic impact, which leads to less aggressive trading on v_1 :*

$$\frac{\partial \tilde{\alpha}_B}{\partial \gamma} = -\frac{3}{(\gamma + 3)^2}. \quad (14)$$

- (iii) *Internalising effect (positive): Fully discretionary firm A exploits the reduced aggressiveness of firm B by trading more aggressively on v_1 :*

$$\frac{\partial \tilde{\alpha}_A}{\partial \gamma} = \frac{3}{2(\gamma + 3)^2}. \quad (15)$$

An increase in the level of quantitative investing (γ) increases both the information processing capacity and the strategic inflexibility of firm B . Higher processing capacity improves access to the additional signal θ_2 , giving firm B a stronger informational advantage in estimating v_2 . The resulting increase in trading aggressiveness on θ_2 is referred to as the *processing capacity effect*. At the same time, greater strategic inflexibility makes firm B more

likely to overestimate the trading aggressiveness of firm A , leading to reduced aggressiveness on v_1 . We refer to this as the *strategic myopia effect*. In response, the fully discretionary firm A increases its aggressiveness on v_1 to exploit firm B 's reduced strategic sensitivity. We refer to this response as the *internalising effect*.

Corollary 1 characterises how these three effects vary with additional signal quality q of the quant department and the level of quantitative investing γ . Figure 2 illustrates three effects for different values of q . Panels (A)-(C) show these effects for $q = 0.04, 0.09$ and 0.12 , respectively. While we choose these values for illustration purposes, they are also realistic because alternative data sources, such as satellite imagery and web crawling data, are typically very noisy and generate signals with low predictive power, corresponding to signal-to-noise ratios broadly in this range.⁴ We combine Figure 2 with Corollary 1 to analyse the net impact of growth in γ on overall trading aggressiveness under different parameter settings.

[Figure 2 here]

Corollary 1. *The magnitude of the processing capacity effect increases with signal quality q . The magnitude of the strategic myopia effect decreases with the level of quantitative investing γ , and it decreases at a faster rate than the internalising effect.*

According to Corollary 1, the positive *processing capacity effect* becomes stronger as signal quality q increases. This is illustrated in Figure 2. In Panel (A), where θ_2 is of low quality ($q = 0.04$), the negative strategic myopia effect dominates the sum of the two positive effects (processing capacity and internalising effects). As a result, increasing the level of quantitative investing γ decreases overall trading aggressiveness for all values of γ . In contrast, Panel (C) shows the case with high-quality θ_2 ($q = 0.12$). Here, the two positive effects dominate the negative strategic myopia effect, so overall trading aggressiveness increases with γ .

⁴ Other main limitations around alternative data sources are incomplete datasets, limited volume, lack of robust data verification, and insufficient historical coverage. See, for example, [Deloitte's report on 'Alternative data for investment decisions: Today's innovation could be tomorrow's requirement'](#) for potential risks of alternative data.

Corollary 1 also shows that the magnitude of the strategic myopia effect decreases as γ increases. This is intuitive because a change from $\gamma = 0$ to a slightly positive value represents a move away from full rationality in firm B , which has a strong marginal impact on how it trades on v_1 . In contrast, the same amount of increase when γ is close to 1 changes little. Moreover, an increase in γ decreases the strategic myopia effect more than it decreases the internalising effect. As a result, for moderate quality θ_2 as shown in Panel (B) of Figure 2, at low γ , the negative effect dominates. But as γ increases, the positive effects catch up and eventually dominate, implying that overall trading aggressiveness first decreases, then increases with γ .

In addition, an increase in signal quality q does not change the strategic myopia effect, but it increases the sum of the processing capacity and internalising effects, without altering their sensitivity to γ . This is because the processing capacity effect is linear in q , while the other two are independent of it.

As trading is the only channel through which information is incorporated into prices in the model, changes in trading aggressiveness naturally translate into changes in price efficiency. Following Kyle (1985) and its extensions (e.g., Subrahmanyam 1991; Holden and Subrahmanyam 1992; Yang and Zhu 2020), we measure price efficiency as $\text{Var}(v|p)^{-1}$. In the baseline equilibrium, where one firm is fully discretionary and the other has a quant department, price efficiency is given by:

$$\text{PE}_{\text{baseline}} = \frac{2\gamma + 6}{(3\gamma - 3q\gamma - q\gamma^2 + 8)\Sigma_0}. \quad (16)$$

Proposition 3 characterises how price efficiency in the baseline equilibrium changes with the growth of quantitative trading.

Proposition 3. *The effect of quantitative investing level γ on price efficiency depends on the quality of the additional signal q obtained by the quant department:*

- (i) *When $0 < q \leq q_1$, price efficiency decreases with the level of quantitative investing γ .*
- (ii) *When $q_1 < q < q_2$, price efficiency decreases with the level of quantitative investing γ if and only if $\gamma < \bar{\gamma}$.*

(iii) When $q \geq q_2$, price efficiency increases with the level of quantitative investing γ .

The impact of quantitative investing on the price efficiency depends on the value of signal quality q obtained by the quant department. When the signal θ_2 is of low quality ($q \leq q_1$), an increase in quantitative investing always harms price efficiency due to the unambiguous drop in overall trading aggressiveness for any γ . When θ_2 is of high quality ($q > q_2$), price efficiency always improves as quantitative investing increases. For moderate signal quality ($q_1 < q < q_2$), the effect is non-monotonic. Price efficiency initially decreases with γ , but increases once γ exceeds the threshold $\bar{\gamma}$, at which point the sum of the processing capacity, strategic myopia, and internalising effects becomes positive. Figure 3 provides a numerical example of the proposition.

[Figure 3 here]

3. Model with two quant firms

We now turn to a setting in which both firms have quant departments. In this extended model, firm A consists of a quant department Q_1 and a discretionary department D_1 , departing from the baseline where it was fully discretionary. Similarly, firm B has a quant department Q_2 and discretionary department D_2 . At $t = 0$, nature determines which department is active in each firm. The probability that firm A 's quant department is active is denoted by $\gamma_1 \in [0, 1]$ and the corresponding probability for firm B is $\gamma_2 \in [0, 1]$.

All traders know that the price function at $t = 1$ takes the form of $p = \lambda'\omega$. The demands of D_1 , Q_1 , D_2 and Q_2 are denoted by x_{D1} , x_{Q1} , x_{D2} , and x_{Q2} , respectively. Since firm A is no longer fully discretionary, both firm A 's and firm B 's ex-ante demand is a weighted sum of demands from their respective quant and discretionary departments:

$$x_A = \gamma_1 x_{Q1} + (1 - \gamma_1) x_{D1}, \quad (17)$$

$$x_B = \gamma_2 x_{Q2} + (1 - \gamma_2) x_{D2}. \quad (18)$$

If present in the market, trader D_2 , who observes v_1 , chooses x_{D2} to maximise expected trading profits as given by

$$E[(v - p(v_1, x_{D2}, x_A))x_{D2}|v_1], \quad (19)$$

and trader Q_2 , who observes v_1 and θ_2 , chooses x_{Q2} to maximise expected trading profits as given by

$$E[(v - p(v_1, x_{Q2}, x_A))x_{Q2}|v_1, \theta_2]. \quad (20)$$

The optimisation problem of D_1 and Q_1 are symmetric to those of D_2 and Q_2 , respectively. The actual demands of D_1 and Q_1 from firm A are given by

$$x_{D1} = \frac{E(v|v_1)}{2\lambda} - \frac{E(x_B|v_1)}{2}, \quad (21)$$

$$x_{Q1} = \frac{E(v | v_1 \theta_2)}{2\lambda} - \frac{E(x_B | v_1, \theta_2)}{2}. \quad (22)$$

However, the strategically myopic quant department Q_2 of firm B incorrectly perceives them as

$$x_{D1} = \frac{E(v|v_1)}{2\lambda}, \quad (23)$$

$$x_{Q1} = \frac{E(v|v_1, \theta_2)}{2\lambda}. \quad (24)$$

As in the baseline model, market makers consider the aggregate order flow ω from firm A , firm B , and noise traders

$$\omega = \gamma_1 x_{Q1} + (1 - \gamma_1) x_{D1} + \gamma_2 x_{Q2} + (1 - \gamma_2) x_{D2} + u. \quad (25)$$

Market makers follow the weak-form efficiency defined by $p = E[v | \omega]$ and set a price that allows them to break even in expectation. The following proposition summarises the equilibrium outcomes in the extended model.

Proposition 4. *When both firms have quant departments, in equilibrium, price is given by $p = \lambda' \omega$, and trading strategies of traders Q_1 , D_1 , Q_2 , and D_2 are given by $x_{Q1} = \alpha_{Q1} v_1 + \beta_1 \theta_2$, $x_{D1} = \alpha_{D1} v_1$, $x_{Q2} = \alpha_{Q2} v_1 + \beta_2 \theta_2$, and $x_{D2} = \alpha_{D2} v_2$, where*

$$\lambda' = \sqrt{\frac{(\gamma_1 + \gamma_2 - 2\gamma_1\gamma_2 + 4)(\gamma_1 + \gamma_2 + 2)}{4(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^2} + q\eta(1 - \eta)}, \quad (26)$$

$$\alpha_{Q1} = \alpha_{Q2} = \frac{1}{4\lambda'}, \quad \alpha_{D1} = \frac{\gamma_1 + 2\gamma_2 - \gamma_1\gamma_2 + 4}{4\lambda'(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)}, \quad (27)$$

$$\beta_{Q1} = \frac{2q - q\gamma_2}{4\lambda'}, \quad \alpha_{D2} = \frac{\gamma_2 + 2\gamma_1 - \gamma_1\gamma_2 + 4}{4\lambda'(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)}, \quad (28)$$

$$\beta_{Q2} = \frac{2q - q\gamma_1}{4\lambda'}, \quad \eta = \frac{\gamma_1 + \gamma_2 - \gamma_1\gamma_2}{2}. \quad (29)$$

Note that, when γ_1 or γ_2 approaches zero, the extended model degenerates to the baseline model. Based on the equilibrium trading strategies defined by Proposition 4, normalised trading aggressiveness of firm A on signals v_1 and θ_2 and of firm B on similar two signals can be derived as

$$\tilde{\alpha}'_A = \frac{3\gamma_2 - 3\gamma_1\gamma_2 + 6}{2(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)}, \quad \tilde{\beta}'_A = \frac{3q\gamma_1(2 - \gamma_2)}{4}, \quad (30)$$

$$\tilde{\alpha}'_B = \frac{3\gamma_1 - 3\gamma_1\gamma_2 + 6}{2(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)}, \quad \tilde{\beta}'_B = \frac{3q\gamma_2(2 - \gamma_1)}{4}. \quad (31)$$

3.1. Growth in quantitative investing by a single firm. We now turn to analysing the *ceteris paribus* impact of an increase in firm B 's quantitative investing level. As before, we define overall trading aggressiveness τ' as the sum of the four normalised trading aggressiveness components in Eqs. (30) and (31). The following proposition characterises how growth in quantitative investing affects overall trading aggressiveness in the extended model where both firms have quant departments.

Proposition 5. *When both firms have quant departments, the impact of quantitative investing level γ_2 of firm B on overall trading aggressiveness τ' depends on four effects:*

- (i) *Processing capacity effect (positive): As firm B relies more on its quant department, it trades more aggressively on the additional signal (θ_2) due to improved information processing capacity:*

$$\frac{\partial \tilde{\beta}'_B}{\partial \gamma_2} = \frac{3q(2 - \gamma_1)}{4}. \quad (32)$$

- (ii) *Strategic myopia effect (negative): The quant department of firm B is less flexible and ignores its strategic impact, which leads to less aggressive trading on v_1 :*

$$\frac{\partial \tilde{\alpha}'_B}{\partial \gamma_2} = -\frac{3\gamma_1 + 3}{(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^2}. \quad (33)$$

- (iii) *Internalising effect (positive): Firm A exploits the strategic myopia of firm B by trading more aggressively on v_1 :*

$$\frac{\partial \tilde{\alpha}'_A}{\partial \gamma_2} = \frac{3 - 3\gamma_1^2}{2(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^2}. \quad (34)$$

- (iv) *Competition effect (negative): As firm B's quant department becomes more active, firm A's quant department trades less aggressively on θ_2 due to increased competition:*

$$\frac{\partial \tilde{\beta}'_A}{\partial \gamma_2} = -\frac{3q\gamma_1}{4}. \quad (35)$$

Proposition 5 suggests that the key results from the baseline model continue to hold when both firms have quant departments. An increase in the quantitative investing level γ_2 leads to: (i) firm B trading more aggressively on θ_2 , as it benefits from improved capacity to process this signal; (ii) firm B trading less aggressively on v_1 , due to greater strategic inflexibility and a higher likelihood of overestimating the opponent's aggressiveness; and (iii) firm A trading more aggressively on v_1 , as its discretionary department internalises firm B's reduced trading aggressiveness.

In addition to these three effects, an increase in γ_2 also reduces firm A's information advantage on θ_2 due to increased competition from firm B. As a result, firm A's quant department trades less aggressively on θ_2 . We refer to this reduction in trading aggressiveness by firm A as the *competition effect*. Corollary 2 analytically describes how these four effects on trading aggressiveness vary with q and γ_2 .

Corollary 2. *When both firms have quant departments, the magnitude of the processing capacity effect increases in signal quality q at a faster rate than the magnitude of the competition effect. The magnitude of the strategic myopia effect decreases in the quantitative investing level γ_2 at a faster rate than the magnitude of the internalising effect.*

Corollary 2 is similar to Corollary 1. The positive processing capacity effect increases more rapidly with q than the negative competition effect does, and the other two effects are independent of q . As a result, when q is high, the processing capacity effect is more likely to dominate, and the net impact of an increase in the quantitative investing level γ_2 on overall trading aggressiveness is positive.

Moreover, the negative strategic myopia effect decreases with γ_2 at a faster rate than the positive internalising effect increases. Therefore, when q is moderate, the strategic myopia effect may dominate at low levels of γ_2 , but as it decreases more steeply with further increases in γ_2 , it no longer dominates. In this case, overall trading aggressiveness becomes non-monotonic in γ_2 . The graphical comparison between the sum of the positive effects and the sum of the absolute values of the negative effects across low, moderate, and high values of q follows a similar pattern to Figure 2.

The changes in trading aggressiveness imply that the effect of increasing γ_2 on price efficiency in the extended model is similar to the effect of increasing γ in the baseline model. The equilibrium price efficiency when both firms have quant departments is given by:

$$\text{PE}_{\text{both quants}} = \frac{4\eta + 6}{(3\gamma_1 + 3\gamma_2 - 2\gamma_1\gamma_2 + 8 - 4q\eta^2 - 6q\eta)\Sigma_0}. \quad (36)$$

Proposition 6 summarises how an increase in quantitative investing by firm B affects price efficiency when both firms have quant departments. Figure 4 illustrates these results.

Proposition 6. *When both firms have quant departments, the effect of quantitative investing level γ_2 of firm B depends on the quality of additional signal q obtained by the quant department:*

- (i) *When $0 < q \leq q_3$, price efficiency decreases with the level of quantitative investing γ_2 .*
- (ii) *When $q_3 < q < q_4$, price efficiency decreases with the level of quantitative investing γ_2 if and only if $\gamma_2 < \bar{\gamma}'$.*
- (iii) *When $q \geq q_4$, price efficiency increases with the level of quantitative investing γ_2 .*

Proposition 6 and Figure 4 show that the main results from the model with one firm with a quant department carry over to the case where both firms have quant departments.

However, the threshold values that classify q as low, moderate, or high differ across the two models. This is because, when both firms have quant departments, the interaction between their strategies alters how signal quality translates into price efficiency.

[Figure 4 here]

Another difference between Figures 3 and 4 is how price efficiency responds to signal quality q when firm B does not have a quant department ($\gamma = 0$ in Figure 3, $\gamma_2 = 0$ in Figure 4). In Figure 3, when $\gamma = 0$, price efficiency remains unchanged as q varies because firm B does not trade on θ_2 and firm A is fully discretionary. In contrast, in Figure 4, price efficiency increases with q even when $\gamma_2 = 0$, since firm A 's quant department still trades on θ_2 , and does so more aggressively when the signal is more precise. This illustrates that signal quality only influences price efficiency when at least one firm uses quant trading strategies.

3.2. Growth in quantitative investing by both firms. In the previous subsection, we examined how an increase in firm B 's quantitative investing level affects trading aggressiveness and price efficiency, holding firm A 's level fixed. We now investigate what happens when both firms increase their quantitative investing levels. To answer this, we analyse the second-order cross-partial derivatives of $\tilde{\alpha}'_A$ (the normalised trading aggressiveness of firm A on signal v_1) and the price efficiency with respect to γ_1 and γ_2 . The results are summarised in the following corollary and illustrated in Figure 5.

Corollary 3. (i) *The internalising effect of an increase in firm B 's quantitative investing level γ_2 decreases as firm A 's quantitative investing level γ_1 increases.* (ii) *The marginal effect of γ_2 on price efficiency decreases as γ_1 increases.*

Recall from the previous analysis that as firm B 's quantitative investing level γ_2 increases, firm A 's discretionary department responds by trading more aggressively on v_1 (an internalising effect), i.e., $\partial \tilde{\alpha}'_A / \partial \gamma_2 > 0$. However, as stated in part (i) of Corollary 3, this internalising effect decreases as γ_1 increases, i.e., $\partial^2 \tilde{\alpha}'_A / \partial \gamma_2 \partial \gamma_1 < 0$ (see Panel A of Figure 5).

[Figure 5 here]

Part (ii) of Corollary 3 further implies that the price efficiency impact of increasing firm B 's quantitative investing level γ_2 is more likely to be negative when firm A 's level γ_1 is high (see Panel B of Figure 5). In this case, the benefits of firm B 's improved information processing are increasingly offset by its reduced strategic responsiveness and by weaker internalisation from firm A . As a result, an increase in γ_2 is more likely to reduce overall trading aggressiveness and harm price efficiency when firm A 's quantitative investing level is high.

To further illustrate the negative impact of increasing firm B 's quantitative investing level γ_2 on price efficiency when firm A is already fully quantitative, Figure 6 plots price efficiency against γ_2 for $\gamma_1 = 1$ and three different levels of signal quality: $q = 0.1$, $q = 0.5$, and $q = 0.9$. The downward-sloping curves indicate that price efficiency is always harmed by increases in firm B 's quantitative investing level γ_2 when the firm A is fully quant.

[Figure 6 here]

One may interpret firm A 's quantitative investing level as a proxy for the overall market's reliance on quantitative strategies, excluding firm B . When this market-level quant intensity is already high, regulators and practitioners should closely monitor further increases in quantitative investing by individual firms. The analytical results and numerical examples in this section suggest that policy interventions aimed at curbing excessive quantitative trading could enhance price efficiency in markets where such activity is already quite high.

4. Conclusion

We theoretically examine the strategic interaction between quantitative and discretionary investors and its impact on market efficiency. From this interaction, we isolate four key effects. First, the *processing capacity effect* reflects quants' superior ability to extract and process information, leading them to trade more aggressively on the additional signal they observe. Second, the *strategic myopia effect* captures quants' limited awareness

of how their trades influence other traders, causing them to trade less aggressively on the public signal. Third, discretionary investors respond strategically by increasing their trading aggressiveness on the public signal, which we term the *internalising effect*. Finally, when multiple quants trade on similar signals, their informational advantage is diluted through a *competition effect*, leading them to trade less aggressively on the additional signal. Together, these forces create a non-monotonic relationship between quantitative participation and market efficiency.

We find that better processing capacity of quants can improve price efficiency, but only when the quality of their private signals is sufficiently strong or their strategic awareness is sufficiently high. In contrast, when quants dominate trading but fail to internalise their market impact, they may crowd into similar positions, trigger feedback loops, and reduce price informativeness. These findings provide a theoretical foundation for empirical patterns observed during episodes such as the Quant Meltdown (e.g., Khandani and Lo 2011; Stein 2009), the May 2010 Flash Crash (e.g., Menkveld and Yueshen 2019), and recent fire-sale spillovers (e.g., Falato et al. 2021), where herding among algorithmic traders led to market instability.

From a policy standpoint, the model suggests that interventions should be sensitive to both the overall level of quantitative investing and the diversity of trading strategies. For example, promoting diversity in how trading signals are processed through incentives for differentiated models, open access to alternative data sources, or regulatory sandbox programs that support the safe testing of novel algorithmic strategies can help reduce herding. Complementing this, circuit breakers that adjust dynamically based on order flow composition, such as triggering earlier when trading is dominated by algorithms, can slow down destabilizing feedback loops. Transparency around models can further support regulators in assessing the risk of coordination failures in algorithmic trading. These measures help mitigate the systemic risks that arise when traders with limited strategic awareness respond simultaneously to similar signals.

For asset managers and market participants, our analysis cautions against over-reliance on signal-based models without accounting for their strategic impact on other investors.

As quantitative strategies become commoditised, their relative advantage can erode not only due to competition, but also due to second-order effects from market instability. Ultimately, our model shows price efficiency is not just about processing more data, but also about understanding how others react to it.

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Figure 1**Model timeline**

This figure illustrates the timeline of the baseline model. At $t = 0$, nature determines which department of firm B participates in the market. At $t = 1$, firms A and B and noise traders trade in the market. At $t = 2$, the security is liquidated.

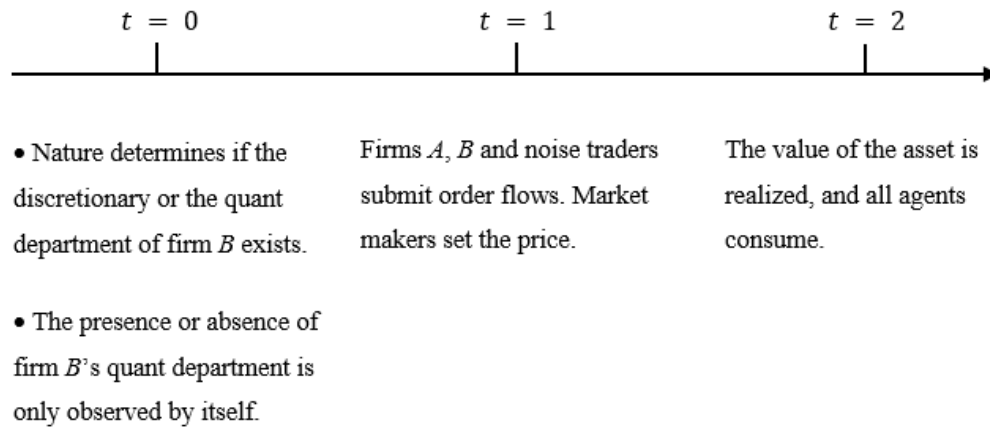
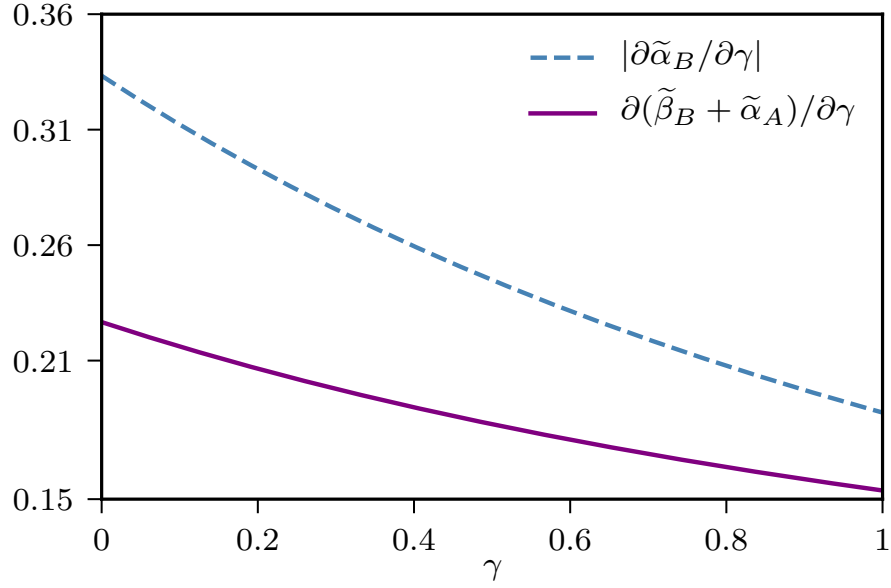
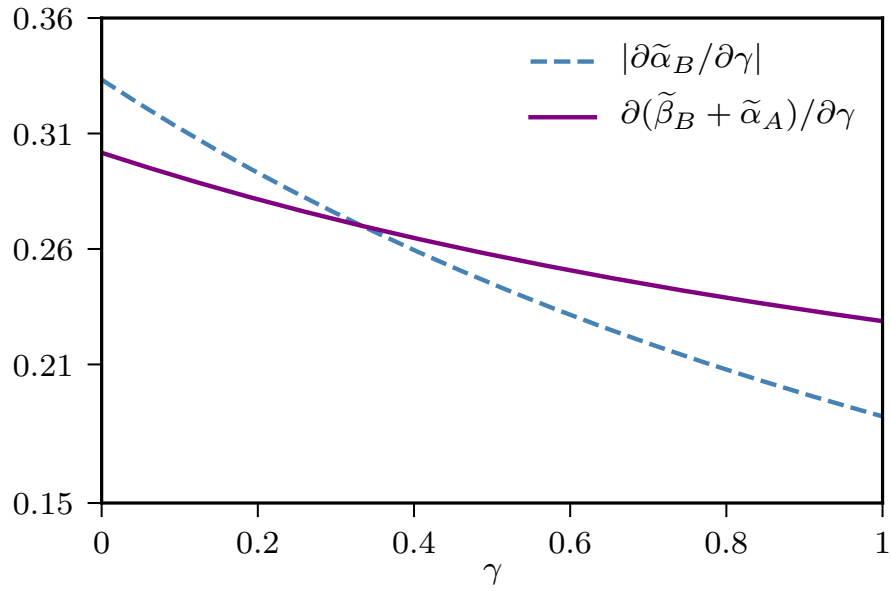


Figure 2**Trading aggressiveness as a function of quantitative investing by firm B (firm A is fully discretionary)**

This figure illustrates the derivatives of trading aggressiveness measures with respect to the level of quantitative investing γ of firm B in the setting of fully discretionary firm A when: (i) q is low; (ii) q is moderate; and (iii) q is high. Panels A, B and C illustrate the derivatives when $q = 0.04$, $q = 0.09$, and $q = 0.12$, respectively. The other parameter value is $\Sigma_0 = 1$. In all panels, the blue dashed line is the absolute value of the strategic myopia effect, and the purple line is the sum of the processing capacity effect and the internalising effect.

(A) $q = 0.04$ (B) $q = 0.09$

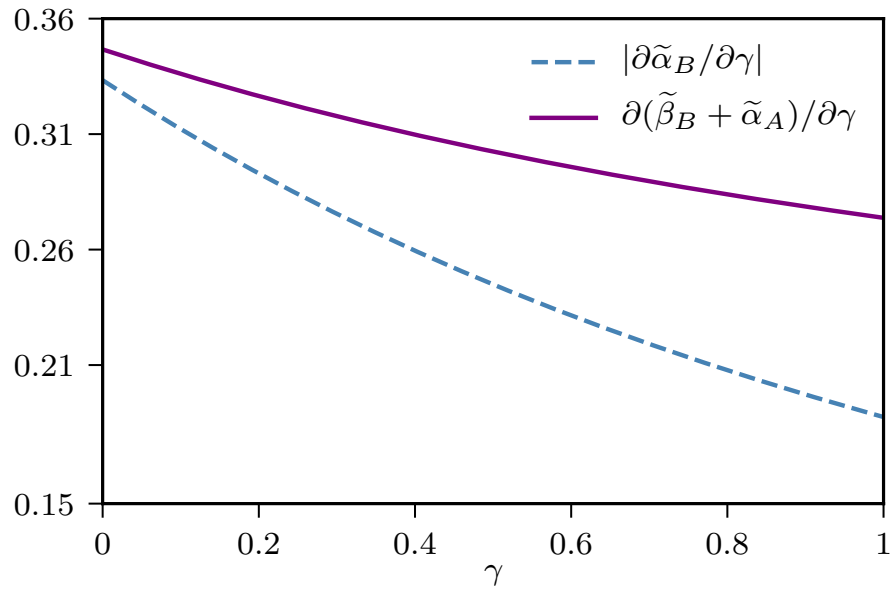
(c) $q = 0.12$

Figure 3**Price efficiency as a function of quantitative investing by firm B (firm A is fully discretionary)**

This figure illustrates the impact of the quantitative investing level γ of firm B on price efficiency in the baseline model with fully discretionary firm A . The figure plots different levels of signal quality q : (i) low ($q = 0.04$), (ii) moderate ($q = 0.09$), and (iii) high ($q = 0.12$). The black dashed line, blue solid line, and grey dashed line plot price efficiency as a function of γ for each of these values, respectively. The other parameter value is $\Sigma_0 = 1$.

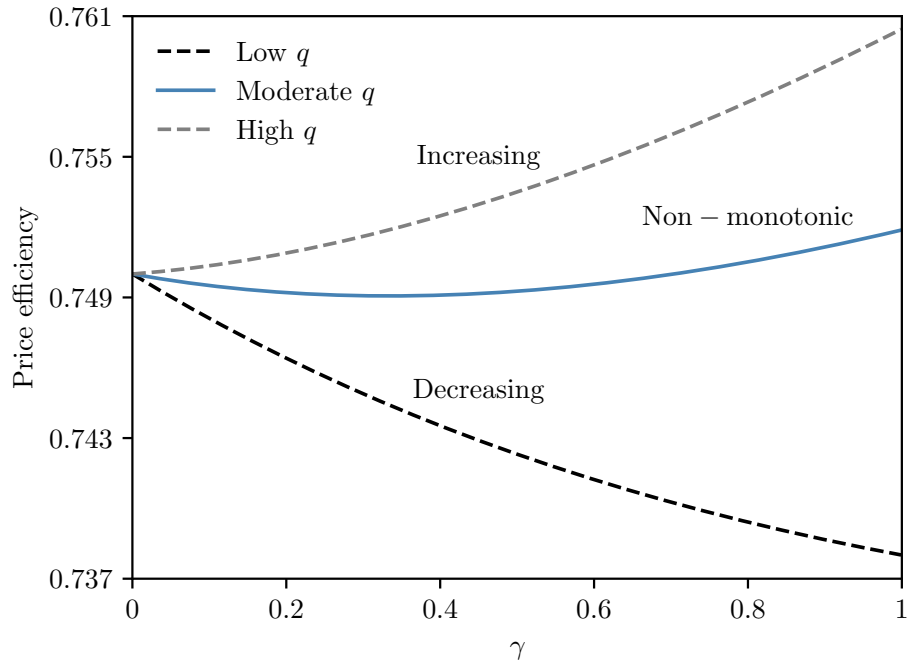


Figure 4**Price efficiency as a function of quantitative investing by firm B (both firms have quant departments)**

This figure illustrates the impact of the quantitative investing level γ_2 of firm B on price efficiency when the quantitative investing level γ_1 of firm A is fixed. The figure plots different levels of signal quality q : (i) low ($q = 0.28$), (ii) moderate ($q = 0.32$), and (iii) high ($q = 0.37$). The black dashed line, blue solid line, and grey dashed line plot price efficiency as a function of γ_2 for each of these values, respectively. The other parameter values are $\Sigma_0 = 1$ and $\gamma_1 = 0.5$.

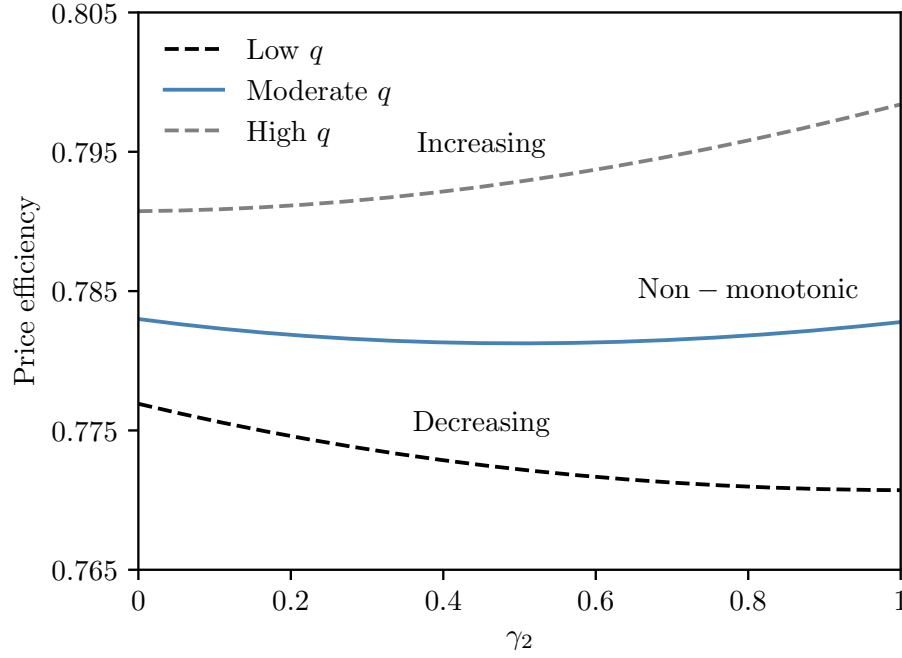
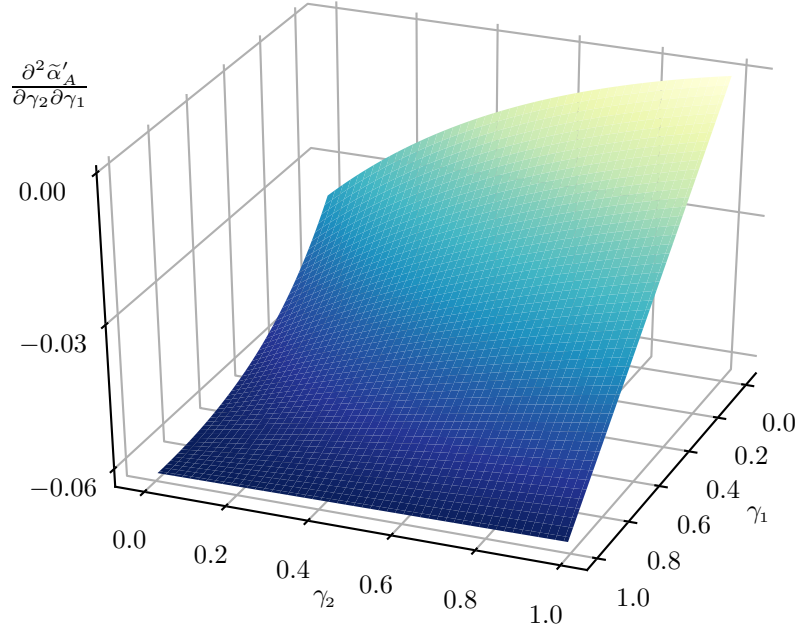


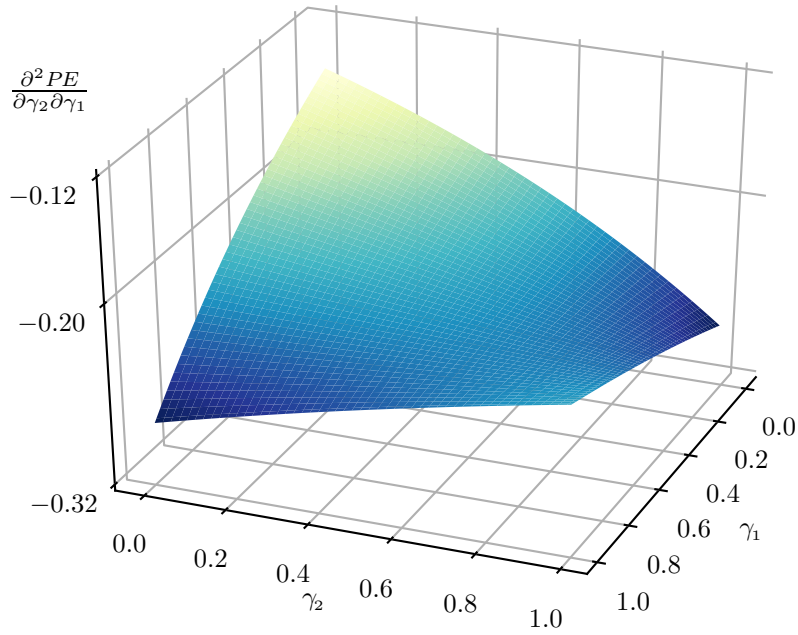
Figure 5

Price efficiency as a function of quantitative investing level by both firms γ_1 and γ_2

The figure illustrates the relationship between price efficiency when both γ_1 and γ_2 vary. Panel (A) plots the second-order partial derivatives of $\tilde{\alpha}'_A$ (the normalised trading aggressiveness of firm A on signal v_1) on γ_2 and γ_1 . Panel (B) plots the second-order partial derivatives of price efficiency on γ_2 and γ_1 . The parameter values are $q = 0.5$ and $\Sigma_0 = 1$.



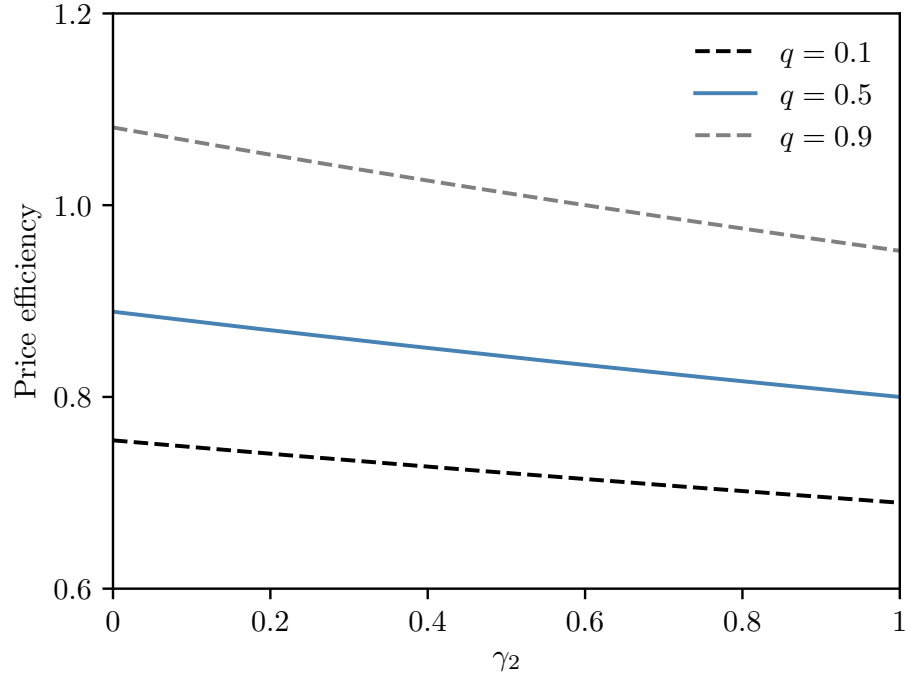
(A) Cross-partial derivative of $\tilde{\alpha}'_A$ on γ_2 and γ_1



(B) Cross-partial derivative of price efficiency on γ_2 and γ_1

Figure 6**Price efficiency as a function of quantitative investing by firm B (firm A is fully quant)**

This figure illustrates the impact of firm B 's quantitative investing level γ_2 on price efficiency when firm A has a quant department but no discretionary department ($\gamma_1 = 1$). Price efficiency is plotted as a function of γ_2 for three values of signal quality: the black dashed line corresponds to $q = 0.1$, the blue line to $q = 0.5$, and the grey dashed line to $q = 0.9$. The other parameter value is $\Sigma_0 = 1$.



Appendix A: Proofs

In this Appendix, we present the proofs of our main results. The proofs of Propositions 1, 2, 3 and Corollary 1 are omitted for brevity, as they follow directly from the proofs of Propositions 4, 5, 6 and Corollary 2, respectively, under the special case where firm A 's quantitative investing level is zero.

Proof of Proposition 4. The quant department of firm B observes $\{v_1, \theta_2\}$ and demands x_{Q2} to maximise the expected profit $E[(v_1 + v_2 - \lambda'x_{Q2} - \lambda'x_A)x_{Q2}|v_1, \theta_2]$. Using the incorrect beliefs of Q_2 about firm A 's trading strategies, as defined in Eqs. (23) and (24), the first-order condition (FOC) yields

$$\begin{aligned} x_{Q2} &= \frac{v_1 + E(v_2|\theta_2)}{2\lambda'} - \frac{E(\gamma_1 x_{Q1}|v_1, \theta_2)}{2} - \frac{E((1 - \gamma_1)x_{D1}|v_1, \theta_2)}{2} \\ &= \frac{v_1}{4\lambda'} + \frac{(2 - \gamma_1)q\theta_2}{4\lambda'}. \end{aligned} \quad (\text{A-1})$$

Thus, the quant department Q_2 follows a linear equilibrium strategy

$$x_{Q2} = \alpha_{Q2}v_1 + \beta_{Q2}\theta_2, \quad (\text{A-2})$$

where the coefficients α_{Q2} and β_{Q2} satisfy

$$\alpha_{Q2} = \frac{1}{4\lambda'} \text{ and } \beta_{Q2} = \frac{2q - q\gamma_1}{4\lambda'}. \quad (\text{A-3})$$

Since the optimisation problems for Q_1 and Q_2 are symmetric, Q_1 's strategy also takes the form

$$x_{Q1} = \alpha_{Q1}v_1 + \beta_{Q1}\theta_2 \quad (\text{A-4})$$

Replacing γ_1 on the right hand side of Eq.(A-3) with γ_2 , α_{Q1} and β_{Q1} satisfy

$$\alpha_{Q1} = \frac{1}{4\lambda'} \text{ and } \beta_{Q1} = \frac{2q - q\gamma_2}{4\lambda'}. \quad (\text{A-5})$$

Now consider the discretionary department of firm B , which observes only v_1 and maximises $E[(v_1 + v_2 - \lambda'x_{D2} - \lambda'x_A)x_{D2}|v_1]$. Taking the FOC of D_2 's profit with respect to x_{D2} and rearranging terms yields

$$\begin{aligned}
x_{D2} &= \frac{v_1}{2\lambda'} - \frac{E(\gamma_1 x_{Q1} | v_1, \theta_2)}{2} - \frac{E((1 - \gamma_1)x_{D1} | v_1, \theta_2)}{2} \\
&= \frac{v_1}{2\lambda'} - \frac{\gamma_1 v_1}{8\lambda'} - \frac{(1 - \gamma_1)x_{D1}}{2}.
\end{aligned} \tag{A-6}$$

By symmetry, the discretionary department of firm A follows

$$x_{D1} = \frac{v_1}{2\lambda'} - \frac{\gamma_2 v_1}{8\lambda'} - \frac{(1 - \gamma_2)x_{D2}}{2}. \tag{A-7}$$

Combining Eqs.(A-6) and (A-7) yields linear equilibrium trading strategies of discretionary departments D_2 and D_1 as

$$x_{D2} = \alpha_{D2}v_1 \text{ and } x_{D1} = \alpha_{D1}v_1, \tag{A-8}$$

where

$$\alpha_{D2} = \frac{(4 + \gamma_2 + 2\gamma_1 - \gamma_1\gamma_2)}{4\lambda'(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)}, \tag{A-9}$$

$$\alpha_{D1} = \frac{(4 + \gamma_1 + 2\gamma_2 - \gamma_2\gamma_1)}{4\lambda'(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)}. \tag{A-10}$$

We now turn to the market maker's decision. The market maker observes the aggregate order flow

$$\omega = \gamma_1 x_{Q1} + (1 - \gamma_1)x_{D1} + \gamma_2 x_{Q2} + (1 - \gamma_2)x_{D2} + u, \tag{A-11}$$

and sets the price $p = E(v|\omega) = \lambda'\omega$. Since $\lambda' = cov(v, \omega)/var(\omega)$, we obtain

$$\lambda' = \frac{(\gamma_1\alpha_{Q1} + (1 - \gamma_1)\alpha_{D1} + \gamma_2\alpha_{Q2} + (1 - \gamma_2)\alpha_{D2})\Sigma_0 + q(\gamma_1\beta_{Q1} + \gamma_2\beta_{Q2})\Sigma_0}{(\gamma_1\alpha_{Q1} + (1 - \gamma_1)\alpha_{D1} + \gamma_2\alpha_{Q2} + (1 - \gamma_2)\alpha_{D2})^2\Sigma_0 + q(\gamma_1\beta_{Q1} + \gamma_2\beta_{Q2})^2\Sigma_0 + \Sigma_0}. \tag{A-12}$$

Multiplying both sides of Eq.(A-12) by λ' and rearranging terms yields

$$\begin{aligned}
&\lambda'^2(\gamma_1\alpha_{Q1} + (1 - \gamma_1)\alpha_{D1} + \gamma_2\alpha_{Q2} + (1 - \gamma_2)\alpha_{D2})^2 + q\lambda'^2(\gamma_1\beta_{Q1} + \gamma_2\beta_{Q2})^2 \\
&+ \lambda'^2 = \lambda'(\gamma_1\alpha_{Q1} + (1 - \gamma_1)\alpha_{D1} + \gamma_2\alpha_{Q2} + (1 - \gamma_2)\alpha_{D2}) + q\lambda'(\gamma_1\beta_{Q1} + \gamma_2\beta_{Q2}).
\end{aligned} \tag{A-13}$$

Substituting Eqs. (A-3), (A-5), (A-9), and (A-10) into Eq. (A-13), we solve explicitly for λ' as

$$\lambda' = \sqrt{\frac{(\gamma_1 + \gamma_2 - 2\gamma_1\gamma_2 + 4)(\gamma_1 + \gamma_2 + 2)}{4(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^2} + q\eta(1 - \eta)}, \quad (\text{A-14})$$

where η is defined as a function of γ_1 and γ_2 as

$$\eta \equiv (\gamma_1 + \gamma_2 - \gamma_1\gamma_2)/2. \quad (\text{A-15})$$

The equilibrium values of $(\alpha_{Q2}, \beta_{Q2}, \alpha_{Q1}, \beta_{Q1}, \alpha_{D2}, \alpha_{D1}, \lambda')$ are thus given by Eqs. (A-3), (A-5), (A-9), (A-10) and (A-14). ■

Proof of Proposition 5. Let $\psi \equiv 1/3\lambda'$. The normalised trading aggressiveness of firm B on signal θ_2 can be calculated as $\tilde{\beta}'_B = \gamma_2\beta_{Q2}/\psi = 3\lambda'\gamma_2\beta_{Q2}$, which simplifies to

$$\tilde{\beta}'_B = 3\lambda'\gamma_2 \cdot \frac{2q - q\gamma_1}{4\lambda'} = \frac{3q\gamma_2(2 - \gamma_1)}{4}. \quad (\text{A-16})$$

Taking the derivative with respect to γ_2 yields

$$\frac{\partial \tilde{\beta}'_B}{\partial \gamma_2} = \frac{3q(2 - \gamma_1)}{4} \cdot \frac{\partial \gamma_2}{\partial \gamma_2} = \frac{3q(2 - \gamma_1)}{4}. \quad (\text{A-17})$$

The normalised trading aggressiveness of firm B on signal v_1 can be calculated as $\tilde{\alpha}'_B = (\gamma_2\alpha_{Q2} + (1 - \gamma_2)\alpha_{D2})/\psi = 3\lambda'\gamma_2\alpha_{Q2} + 3\lambda'(1 - \gamma_2)\alpha_{D2}$, which simplifies to

$$\begin{aligned} \tilde{\alpha}'_B &= 3\lambda'\gamma_2 \cdot \frac{1}{4\lambda'} + 3\lambda'(1 - \gamma_2) \cdot \frac{\gamma_2 + 2\gamma_1 - \gamma_1\gamma_2 + 4}{4\lambda'(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)} \\ &= \frac{3\gamma_1 - 3\gamma_1\gamma_2 + 6}{2(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)}. \end{aligned} \quad (\text{A-18})$$

Taking the derivative with respect to γ_2 yields

$$\begin{aligned} \frac{\partial \tilde{\alpha}'_B}{\partial \gamma_2} &= \frac{3((\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)(-\gamma_1) - (\gamma_1 - \gamma_1\gamma_2 + 2)(1 - \gamma_1))}{2(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^2} \\ &= -\frac{3\gamma_1 + 3}{(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^2}. \end{aligned} \quad (\text{A-19})$$

The normalised trading aggressiveness of firm A on signal θ_2 can be calculated as $\tilde{\beta}'_A = \gamma_1\beta_{Q1}/\psi = 3\lambda'\gamma_1\beta_{Q1}$, which simplifies to

$$\tilde{\beta}'_A = 3\lambda'\gamma_1 \cdot \frac{2q - q\gamma_2}{4\lambda'} = \frac{3q\gamma_1(2 - \gamma_2)}{4}. \quad (\text{A-20})$$

Taking the derivative with respect to γ_2 yields

$$\frac{\partial \tilde{\beta}'_A}{\partial \gamma_2} = \frac{3q\gamma_1}{4} \cdot \frac{\partial(2 - \gamma_2)}{\partial \gamma_2} = -\frac{3q\gamma_1}{4}. \quad (\text{A-21})$$

The normalised trading aggressiveness of firm A on signal v_1 can be calculated as $\tilde{\alpha}'_A = (\gamma_1\alpha_{Q1} + (1 - \gamma_1)\alpha_{D1})/\psi = 3\lambda'\gamma_1\alpha_{Q1} + 3\lambda'(1 - \gamma_1)\alpha_{D1}$, which simplifies to

$$\begin{aligned} \tilde{\alpha}'_A &= 3\lambda'\gamma_1 \cdot \frac{1}{4\lambda'} + 3\lambda'(1 - \gamma_1) \cdot \frac{\gamma_1 + 2\gamma_2 - \gamma_1\gamma_2 + 4}{4\lambda'(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)} \\ &= \frac{3\gamma_2 - 3\gamma_1\gamma_2 + 6}{2(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)}. \end{aligned} \quad (\text{A-22})$$

Taking the derivative with respect to γ_2 yields

$$\begin{aligned} \frac{\partial \tilde{\alpha}'_A}{\partial \gamma_2} &= \frac{3((\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)(1 - \gamma_1) - (\gamma_2 - \gamma_1\gamma_2 + 2)(1 - \gamma_1))}{2(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^2} \\ &= \frac{3 - 3\gamma_1^2}{2(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^2}. \end{aligned} \quad (\text{A-23})$$

Therefore, for $\gamma_1 \in [0, 1]$ and $\gamma_2 \in [0, 1]$, we have $\partial \tilde{\beta}'_B / \partial \gamma_2 > 0$, $\partial \tilde{\alpha}'_B / \partial \gamma_2 < 0$, $\partial \tilde{\beta}'_A / \partial \gamma_2 \leq 0$ (the equal sign holds only when $\gamma_1 = 0$), $\partial \tilde{\alpha}'_A / \partial \gamma_2 \geq 0$ (the equal sign holds only when $\gamma_1 = 1$). ■

Proof of Corollary 2. We examine how the cross-partial derivatives reveal the relative sensitivities of each effect to changes in signal quality q and quantitative investing level γ_2 .

(i) *Processing capacity vs. competition effect:* We take the partial derivative of the processing capacity effect in Eq. (A-17)

$$\frac{\partial^2 \tilde{\beta}'_B}{\partial \gamma_2 \partial q} = \frac{3(2 - \gamma_1)}{4}. \quad (\text{A-24})$$

We have $\partial^2 \tilde{\beta}'_B / \partial \gamma_2 \partial q > 0$, which combined with $\partial \tilde{\beta}'_B / \partial \gamma_2 > 0$, means that the magnitude of the the processing capacity effect is intensified by increased signal quality q . Similarly, taking the derivative of the competition effect in Eq. (A-21) with respect to the signal

quality q obtains

$$\frac{\partial^2 \tilde{\beta}'_A}{\partial \gamma_2 \partial q} = -\frac{3\gamma_1}{4}. \quad (\text{A-25})$$

We have $\partial^2 \tilde{\beta}'_A / \partial \gamma_2 \partial q \leq 0$, which combined with $\partial \tilde{\beta}'_A / \partial \gamma_2 < 0$, means that the magnitude of the competition effect is intensified by increased signal quality q . Comparing the two magnitudes

$$\frac{\partial^2 \tilde{\beta}'_B}{\partial \gamma_2 \partial q} - \left| \frac{\partial^2 \tilde{\beta}'_A}{\partial \gamma_2 \partial q} \right| = \frac{3}{2} \quad (\text{A-26})$$

confirms that the processing capacity effect increases in q at a faster rate than the competition effect does.

(ii) *Strategic myopia vs. internalising effect*: Taking the derivative of the strategic myopia effect in Eq. (A-19) with respect to the quantitative investing level γ_2 yields

$$\begin{aligned} \frac{\partial^2 \tilde{\alpha}'_B}{\partial \gamma_2 \partial \gamma_2} &= -(3 + 3\gamma_1) \cdot \left(-\frac{2}{(3 + \gamma_1 + \gamma_2 - \gamma_1\gamma_2)^3} \right) \cdot \frac{\partial(3 + \gamma_1 + \gamma_2 - \gamma_1\gamma_2)}{\partial \gamma_2} \\ &= \frac{6(1 + \gamma_1)(1 - \gamma_1)}{(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^3}. \end{aligned} \quad (\text{A-27})$$

Since $\partial^2 \tilde{\alpha}'_B / \partial \gamma_2 \partial \gamma_2 \geq 0$ and $\partial \tilde{\alpha}'_B / \partial \gamma_2 < 0$, the magnitude of the strategic myopia effect is weakened in the quantitative investing level γ_2 . Similarly, taking the derivative of the internalising effect in Eq. (A-23) with respect to the quantitative investing level γ_2 yields

$$\begin{aligned} \frac{\partial^2 \tilde{\alpha}'_A}{\partial \gamma_2 \partial \gamma_2} &= \frac{(3 - 3\gamma_1^2)}{2} \cdot \frac{-2(1 - \gamma_1)}{(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^2} \\ &= -\frac{3(1 - \gamma_1)^2(1 + \gamma_1)}{(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^2}. \end{aligned} \quad (\text{A-28})$$

Since $\partial^2 \tilde{\alpha}'_A / \partial \gamma_2 \partial \gamma_2 \leq 0$ and $\partial \tilde{\alpha}'_A / \partial \gamma_2 \geq 0$, the magnitude of the internalising effect is weakened by the increased quantitative investing level γ_2 . Comparing Eqs.(A-27) and (A-28)

$$\frac{\partial^2 \tilde{\alpha}'_B}{\partial \gamma_2 \partial \gamma_2} - \left| \frac{\partial^2 \tilde{\alpha}'_A}{\partial \gamma_2 \partial \gamma_2} \right| = \frac{3(1 + \gamma_1)^2(1 - \gamma_1)}{(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^3} > 0 \quad (\text{A-29})$$

confirms the magnitude of the strategic myopia effect decreases in quantitative investing level γ_2 at a faster rate than the internalising effect does.

■

Proof of Proposition 6. The covariance between the fundamental value v and the price p can be calculated as

$$\begin{aligned} \text{cov}(v, p) &= \text{cov}(v_1 + v_2, \lambda' \gamma_1 \alpha_{Q1} v_1 + \lambda' (1 - \gamma_1) \alpha_{D1} v_1 + \lambda' \gamma_2 \alpha_{Q2} v_1 + \\ &\quad \lambda' \gamma_1 \beta_{Q1} \theta_2 + \lambda' \gamma_2 \beta_{Q2} \theta_2) \\ &= \frac{\gamma_1 + \gamma_2 - 2\gamma_1\gamma_2 + 4}{2(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)} \Sigma_0 + \frac{q(\gamma_1 + \gamma_2 - \gamma_1\gamma_2)}{2} \Sigma_0. \end{aligned} \quad (\text{A-30})$$

Because $\lambda = \text{cov}(v, \omega) / \text{var}(\omega)$, we get $\text{cov}(v, p) = \text{var}(p)$. Using the projection theorem, the conditional variance $\text{var}(v|p)$ is

$$\begin{aligned} \text{var}(v|p) &= \text{var}(v) - \frac{\text{cov}(v, p)^2}{\text{var}(p)} \\ &= 2\Sigma_0 - \frac{\gamma_1 + \gamma_2 - 2\gamma_1\gamma_2 + 4}{2(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)} \Sigma_0 - \frac{q(\gamma_1 + \gamma_2 - \gamma_1\gamma_2)}{2} \Sigma_0. \end{aligned} \quad (\text{A-31})$$

The first-order derivative of $\text{var}(v|p)^{-1} = \text{PE}_{\text{both quants}}$ with respect to the quantitative investing level γ_2 is

$$\begin{aligned} \frac{\partial \text{PE}_{\text{both quants}}}{\partial \gamma_2} &= -\frac{1}{\text{var}(v_1 + v_2|p)^2} \cdot \frac{\partial \text{var}(v|p)}{\partial \gamma_2} \\ &= -\frac{1}{\text{var}(v_1 + v_2|p)^2} \left(\frac{(1 + \gamma_1)^2}{2(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^2} + q\left(\frac{\gamma_1}{2} - \frac{1}{2}\right) \right) \Sigma_0^{-1} \\ &\propto -\left(\frac{(1 + \gamma_1)^2}{2(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^2} + q\left(\frac{\gamma_1}{2} - \frac{1}{2}\right) \right). \end{aligned} \quad (\text{A-32})$$

To determine the sign of Eq. (A-32) we discuss the following two scenarios: $\gamma_1 = 1$ and $0 \leq \gamma_1 < 1$. (i) If $\gamma_1 = 1$ (firm A is fully quantitative), we have the following relationships:

$$\frac{(1 + \gamma_1)^2}{2(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^2} > 0 \quad \text{and} \quad \frac{\gamma_1}{2} - \frac{1}{2} = 0. \quad (\text{A-33})$$

Substituting Eq. (A-33) into Eq. (A-32) yields $\partial \text{PE}_{\text{both quants}} / \partial \gamma_2 < 0$ for any γ_2 . That is, price efficiency always decreases with the quantitative investing level γ_2 when $\gamma_1 = 1$.

(ii) If $0 \leq \gamma_1 < 1$ and firm A has a discretionary department, $\partial \text{PE}_{\text{both quants}} / \partial \gamma_2 > 0$ holds if and only if

$$\begin{aligned}
& - \left(\frac{(1 + \gamma_1)^2}{2(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^2} + q \left(\frac{\gamma_1}{2} - \frac{1}{2} \right) \right) > 0 \\
& \iff \gamma_2 > \frac{1 + \gamma_1}{(1 - \gamma_1)\sqrt{q(1 - \gamma_1)}} - \frac{3 + \gamma_1}{1 - \gamma_1} \equiv \bar{\gamma}'.
\end{aligned} \tag{A-34}$$

Note that for $\bar{\gamma}'$ to lie inside of $(0, 1)$, the value of q must satisfy the following inequality

$$\begin{aligned}
0 & < \frac{1 + \gamma_1}{(1 - \gamma_1)\sqrt{q(1 - \gamma_1)}} - \frac{3 + \gamma_1}{1 - \gamma_1} < 1 \\
& \iff q > \frac{(1 + \gamma_1)^2}{16(1 - \gamma_1)} \equiv q_3 \quad \text{and} \quad q < \frac{(1 + \gamma_1)^2}{(3 + \gamma_1)^2(1 - \gamma_1)} \equiv q_4.
\end{aligned} \tag{A-35}$$

Therefore, when $q_3 < q < q_4$, price efficiency decreases (resp. increases) with the quantitative investing level γ_2 if γ_2 is below (resp. above) the threshold $\bar{\gamma}'$. When $q < q_3$, we have $\bar{\gamma}' > 1$, which means $\text{PE}_{\text{both quants}}/\partial\gamma_2 < 0$ for any γ_2 . When $q = q_3$, we have $\bar{\gamma}' = 1$ and thus $\gamma_2 \leq \bar{\gamma}'$ and $\partial\text{PE}_{\text{both quants}}/\partial\gamma_2 \leq 0$, of which the equal signs hold for $\gamma_2 = 1$. Consequently, when $0 < q \leq q_3$, price efficiency decreases with γ_2 .

When $q > q_4$, we have $\bar{\gamma}' < 0$, which means $\partial\text{PE}_{\text{both quants}}/\partial\gamma_2 > 0$ for any γ_2 . When $q = q_4$, we have $\bar{\gamma}' = 0$ and thus $\gamma_2 \leq \bar{\gamma}'$ and $\partial\text{PE}_{\text{both quants}}/\partial\gamma_2 \leq 0$, of which the equal signs hold for $\gamma_2 = 0$. Consequently, when $q \geq q_4$, price efficiency increases with γ_2 .

Substituting $\gamma_1 = 0$ into Eq. (A-35), we obtain the values of q_1 and q_2 in Proposition (3) as

$$q_1 \equiv \frac{1}{16} \quad \text{and} \quad q_2 \equiv \frac{1}{9}. \tag{A-36}$$

■

Proof of Corollary 3. We determine the sign of the second-order cross partial derivative of $\tilde{\alpha}'_A$ with respect to γ_1 and γ_2 as follows

$$\begin{aligned}
\frac{\partial^2 \tilde{\alpha}'_A}{\partial\gamma_2 \partial\gamma_1} &= \frac{3}{2}(1 - \gamma_1^2) \cdot \left(- \frac{2(1 - \gamma_1)}{(3 + \gamma_1 + \gamma_2 - \gamma_1\gamma_2)^3} \right) \\
&= - \frac{3(1 - \gamma_1)(1 - \gamma_1^2)}{(3 + \gamma_1 + \gamma_2 - \gamma_1\gamma_2)^3} < 0.
\end{aligned} \tag{A-37}$$

The second-order cross partial derivative of price efficiency satisfies

$$\begin{aligned} \frac{\partial \text{PE}_{\text{both quants}}}{\partial \gamma_2 \partial \gamma_1} &= \frac{2}{\text{var}(v|p)^3} \cdot \frac{\partial \text{var}(v|p)}{\partial \gamma_1} \cdot \frac{\partial \text{var}(v|p)}{\partial \gamma_2} - \frac{1}{\text{var}(v|p)^2} \cdot \frac{\partial \text{var}(v|p)}{\partial \gamma_2 \partial \gamma_1} \\ &\propto \frac{2}{\text{var}(v|p)} \cdot \frac{\partial \text{var}(v|p)}{\partial \gamma_1} \cdot \frac{\partial \text{var}(v|p)}{\partial \gamma_2} - \frac{\partial \text{var}(v|p)}{\partial \gamma_2 \partial \gamma_1}. \end{aligned} \quad (\text{A-38})$$

We can derive that $\frac{2}{3}\Sigma_0^{-1} \leq \text{var}(v|p)^{-1} \leq \frac{8}{7}\Sigma_0^{-1}$. Taking first-order derivatives of conditional variance $\text{var}(v|p)$ with respect to γ_1 and γ_2 obtains

$$\frac{\partial \text{var}(v|p)}{\partial \gamma_1} = \left(\frac{(1 + \gamma_2)^2}{2(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^2} + q \left(\frac{\gamma_2}{2} - \frac{1}{2} \right) \right) \Sigma_0, \quad (\text{A-39})$$

$$\frac{\partial \text{var}(v|p)}{\partial \gamma_2} = \left(\frac{(1 + \gamma_1)^2}{2(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^2} + q \left(\frac{\gamma_1}{2} - \frac{1}{2} \right) \right) \Sigma_0. \quad (\text{A-40})$$

From Eqs. (A-39) and (A-40), we can determine the value range of $\partial \text{var}(v|p)/\partial \gamma_1$ and $\partial \text{var}(v|p)/\partial \gamma_2$ as $-\frac{15}{32}\Sigma_0 \leq \partial \text{var}(v|p)/\partial \gamma_1 \leq \frac{1}{8}\Sigma_0$ and $-\frac{15}{32}\Sigma_0 \leq \partial \text{var}(v|p)/\partial \gamma_2 \leq \frac{1}{8}\Sigma_0$. The second-order cross partial derivative of $\text{var}(v|p)$ is

$$\frac{\partial^2 \text{var}(v|p)}{\partial \gamma_2 \partial \gamma_1} = \left(\frac{q}{2} + \frac{2(1 + \gamma_1)(1 + \gamma_2)}{(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^3} \right) \Sigma_0. \quad (\text{A-41})$$

To determine the sign of the second-order cross partial derivative of price efficiency measure, we analyse the following five scenarios case by case:

- (1) $\partial \text{var}(v|p)/\partial \gamma_1 < 0$ and $\partial \text{var}(v|p)/\partial \gamma_2 > 0$,
- (2) $\partial \text{var}(v|p)/\partial \gamma_1 < 0$ and $\partial \text{var}(v|p)/\partial \gamma_2 < 0$,
- (3) $\partial \text{var}(v|p)/\partial \gamma_1 > 0$ and $\partial \text{var}(v|p)/\partial \gamma_2 < 0$,
- (4) $\partial \text{var}(v|p)/\partial \gamma_1 > 0$ and $\partial \text{var}(v|p)/\partial \gamma_2 > 0$,
- (5) $\partial \text{var}(v|p)/\partial \gamma_1$ or $\partial \text{var}(v|p)/\partial \gamma_2 = 0$.

(1) If $\partial \text{var}(v|p)/\partial \gamma_1 < 0$ and $\partial \text{var}(v|p)/\partial \gamma_2 > 0$, as $2(\text{var}(v|p))^{-1} > 0$, $-\partial \text{var}(v|p)/\partial \gamma_1 \partial \gamma_2 < 0$, and $(\partial \text{var}(v|p)/\partial \gamma_1)(\partial \text{var}(v|p)/\partial \gamma_2) < 0$, we obtain that $\partial \text{PE}_{\text{both quants}}/\partial \gamma_2 \partial \gamma_1$ is negative.

(2) If $\partial \text{var}(v|p)/\partial \gamma_1 < 0$ and $\partial \text{var}(v|p)/\partial \gamma_2 < 0$, since $\frac{2}{3}\Sigma_0^{-1} \leq \text{var}(v|p)^{-1} \leq \frac{8}{7}\Sigma_0^{-1}$ and $0 < -2\partial \text{var}(v|p)/\partial \gamma_1 \leq \frac{15}{16}\Sigma_0$, we have

$$\begin{aligned}
& \frac{2}{\text{var}(v|p)} \cdot \frac{\partial \text{var}(v|p)}{\partial \gamma_1} \cdot \frac{\partial \text{var}(v|p)}{\partial \gamma_2} - \frac{\partial \text{var}(v|p)}{\partial \gamma_2 \partial \gamma_1} \\
& \leq -\frac{15}{14} \frac{\partial \text{var}(v|p)}{\partial \gamma_2} - \frac{\partial \text{var}(v|p)}{\partial \gamma_1 \partial \gamma_2} \\
& = -\frac{15}{14} \left(\frac{(1+\gamma_1)^2}{2(\gamma_1 + \gamma_2 - \gamma_1 \gamma_2 + 3)^2} + \frac{q\gamma_1 - q}{2} \right) \Sigma_0 \\
& \quad - \left(\frac{q}{2} + \frac{2\gamma_1 + 2\gamma_2 + 2\gamma_1 \gamma_2 + 2}{(\gamma_1 + \gamma_2 - \gamma_1 \gamma_2 + 3)^3} \right) \Sigma_0 \\
& \equiv s.
\end{aligned} \tag{A-42}$$

From Eq.(A-42), $\partial s / \partial q = (1 - 15\gamma_1)\Sigma_0 / 28$. If $\gamma_1 \geq 1/15$, we have $\partial s / \partial q \leq 0$. Hence,

$$\begin{aligned}
s & \leq s|_{q=0} = -\frac{(1+\gamma_1)(60\gamma_1 + 71\gamma_2 - 15\gamma_1^2\gamma_2 + 15\gamma_1^2 + 101)}{28(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^3} \Sigma_0 \\
& < -\frac{\Sigma_0}{28} < 0.
\end{aligned} \tag{A-43}$$

If $\gamma_1 < 1/15$, we have $\partial s / \partial q > 0$ and thus the following inequality

$$\begin{aligned}
s & \leq s|_{q=1} = -\frac{(1+\gamma_1)(60\gamma_1 + 71\gamma_2 - 15\gamma_1^2\gamma_2 + 15\gamma_1^2 + 101)}{28(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^3} \Sigma_0 + \left(\frac{1}{28} - \frac{15\gamma_1}{28} \right) \Sigma_0 \\
& < -\frac{1}{28} \Sigma_0 + \frac{1}{28} \Sigma_0 - \frac{15\gamma_1}{28} \Sigma_0 < 0.
\end{aligned} \tag{A-44}$$

According to Eqs. (A-43) and (A-44), we know that $\partial \text{PE}_{\text{both quants}} / \partial \gamma_2 \gamma_1$ is negative when $\partial \text{var}(v|p) / \partial \gamma_1 < 0$ and $\partial \text{var}(v|p) / \partial \gamma_2 < 0$.

(3) If $\partial \text{var}(v|p) / \partial \gamma_1 > 0$ and $\partial \text{var}(v|p) / \partial \gamma_2 < 0$, as $2(\text{var}(v|p))^{-1} > 0$, $-\partial \text{var}(v|p) / \partial \gamma_1 \partial \gamma_2 < 0$, and $(\partial \text{var}(v|p) / \partial \gamma_1)(\partial \text{var}(v|p) / \partial \gamma_2) < 0$, we obtain that $\partial \text{PE}_{\text{both quants}} / \partial \gamma_2 \gamma_1$ is negative.

(4) If $\partial \text{var}(v|p) / \partial \gamma_1 > 0$ and $\partial \text{var}(v|p) / \partial \gamma_2 > 0$, as $\frac{4}{3}\Sigma_0^{-1} \leq 2(\text{var}(v|p))^{-1} \leq \frac{16}{7}\Sigma_0^{-1}$ and $0 < (\partial \text{var}(v|p) / \partial \gamma_1)(\partial \text{var}(v|p) / \partial \gamma_2) \leq \frac{1}{64}\Sigma_0^2$, we obtain

$$\begin{aligned}
& \frac{2}{\text{var}(v|p)} \cdot \frac{\partial \text{var}(v|p)}{\partial \gamma_1} \cdot \frac{\partial \text{var}(v|p)}{\partial \gamma_2} - \frac{\partial \text{var}(v|p)}{\partial \gamma_2 \partial \gamma_1} \\
& \leq \frac{1}{28} \Sigma_0 - \left(\frac{q}{2} + \frac{2\gamma_1 + 2\gamma_2 + 2\gamma_1\gamma_2 + 2}{(\gamma_1 + \gamma_2 - \gamma_1\gamma_2 + 3)^3} \right) \Sigma_0 \\
& < \frac{1}{28} \Sigma_0 - \frac{1}{16} \Sigma_0 < 0.
\end{aligned} \tag{A-45}$$

(5) If $\partial \text{var}(v|p)/\partial \gamma_1 = 0$ or $\partial \text{var}(v|p)/\partial \gamma_2 = 0$, we have $\partial PE_{\text{both quants}}/\partial \gamma_2 \partial \gamma_1 = -(\text{var}(v|p))^{-2} \partial \text{var}(v|p)/\partial \gamma_2 \partial \gamma_1 < 0$.

Hence, $\partial PE_{\text{both quants}}/\partial \gamma_2 \partial \gamma_1 < 0$ for all combinations of γ_1 , γ_2 , and q .

■

Appendix B: Partially myopic quant

In the baseline model, firm A is fully discretionary, and firm B has a quant department that is assumed to have complete strategic myopia. That is, it entirely ignores the impact of its trading on firm A 's behavior. In this appendix, we relax this assumption by allowing for partial strategic responsiveness. Specifically, for proportion h of the quant department Q 's orders, Q incorrectly ignores that the fully discretionary firm A reacts to the corresponding price impact by reducing trading aggressiveness, where $0 < h \leq 1$. Therefore, h captures the degree of strategic inflexibility or myopia within the quant department, ranging from full strategic awareness when $h = 0$ to complete strategic myopia when $h = 1$.

Let the market maker's pricing rule be $p = \lambda^* \omega$. Under partial strategic responsiveness, the quant department now believes that firm A 's demand is

$$x_A = \frac{E(v \mid \phi_A)}{2\lambda^*} - \frac{(1-h)E(x_B \mid \phi_A)}{2}. \quad (\text{B-1})$$

All other features of the baseline model remain unchanged. The following proposition characterises the equilibrium in this extended model with a partially myopic quant department.

Proposition B1. *When the quant department is partially myopic, the equilibrium pricing rule of market makers is characterised by $p = \lambda^* \omega$, and equilibrium trading strategies of traders A , Q , and D are given by $x_A = \alpha_A^* v_1$, $x_Q = \alpha_Q^* v_1 + \beta_Q^* \theta_2$, and $x_D = \alpha_D^* v_1$, where*

$$\lambda^* = \sqrt{\frac{(\gamma h + 4)(\gamma h + 2)}{4(\gamma h + 3)^2} + \frac{n\gamma(2 - \gamma)}{4}}, \quad (\text{B-2})$$

$$\alpha_Q^* = \frac{\gamma h - h + 4}{4\lambda^*(\gamma h + 3)}, \quad \alpha_D^* = \frac{\gamma h + 4}{4\lambda^*(\gamma h + 3)}, \quad (\text{B-3})$$

$$\beta_Q^* = \frac{q}{2\lambda^*}, \quad \alpha_A^* = \frac{\gamma h + 2}{2\lambda^*(\gamma h + 3)}. \quad (\text{B-4})$$

Proofs of all results in this appendix are omitted for brevity. They closely follow the derivations in the baseline model and can be provided upon request.

The normalised trading aggressiveness measures of firm A on signal v_1 , and of firm B on signals v_1 and θ_2 are derived as

$$\tilde{\alpha}_A^* = \frac{3\gamma h + 6}{2\gamma h + 6}, \quad \tilde{\alpha}_B^* = \frac{3}{\gamma h + 3}, \quad \tilde{\beta}_B^* = \frac{3q\gamma}{2}. \quad (\text{B-5})$$

Define the overall trading aggressiveness as $\tau^* \equiv \tilde{\alpha}_A^* + \tilde{\alpha}_B^* + \tilde{\beta}_B^*$. The following proposition shows the effects of growth in quantitative investing on τ^* .

Proposition B2. *Assume the quant department of firm B ignores its price impact with probability $h \in (0, 1]$. When quantitative investing level γ increases, the impact on overall trading aggressiveness τ depends on three effects:*

- (i) *Processing capacity effect (positive): As firm B relies more on its quant department, it trades more aggressively on the additional signal (θ_2) due to improved information processing capacity:*

$$\frac{\partial \tilde{\beta}_B^*}{\partial \gamma} = \frac{3q}{2}. \quad (\text{B-6})$$

- (ii) *Strategic myopia effect (negative): The quant department of firm B is less flexible and ignores its strategic impact, which leads to less aggressive trading on v_1 :*

$$\frac{\partial \tilde{\alpha}_B^*}{\partial \gamma} = -\frac{3h}{(\gamma h + 3)^2}. \quad (\text{B-7})$$

- (iii) *Internalising effect (positive): Fully discretionary firm A exploits the reduced aggressiveness of firm B by trading more aggressively on v_1 :*

$$\frac{\partial \tilde{\alpha}_A^*}{\partial \gamma} = \frac{3h}{2(\gamma h + 3)^2}. \quad (\text{B-8})$$

Proposition B2 shows that the main insights from the baseline model remain valid under partially responsive quant department. Specifically, an increase in the quantitative investing level γ continues to generate three distinct effects: a positive processing capacity effect, a negative strategic myopia effect, and a positive internalising effect. Importantly, the strategic myopia effect, $\partial \tilde{\alpha}_B^* / \partial \gamma$, and the internalising effect, $\partial \tilde{\alpha}_A^* / \partial \gamma$, are both independent of signal quality q and vanish as the degree of strategic responsiveness h approaches

zero. In contrast, the processing capacity effect, $\partial \tilde{\beta}_B^* / \partial \gamma$, is independent of h and remains constant regardless of the degree of strategic responsiveness.

Corollary B1. *When the quant department of firm B ignores its price impact with probability h :*

- (i) *The magnitude of the processing capacity effect increases with signal quality q .*
- (ii) *The magnitude of the strategic myopia effect decreases with the quantitative investing level γ at a faster rate than the internalising effect.*
- (iii) *The magnitude of the strategic myopia effect increases with h at a faster rate than the internalising effect.*

Corollary B1 describes how the three main effects (processing capacity, strategic myopia, and internalising effects) change with signal quality q , quantitative investing level γ , and the degree of strategic myopia h . Two key insights follow. First, the relationships of these effects with respect to q and γ remain qualitatively consistent with Corollary 1. Second, increasing h amplifies the strategic myopia effect more than the internalising effect. The intuition is that a more myopic quant department (higher h) contributes more to strategic inflexibility, causing a sharper decline in firm B 's trading aggressiveness on v_1 .

As a result, the overall impact of increasing γ on trading aggressiveness is more likely to be positive when q is high or h is low (processing capacity dominates), and more likely to be negative when q is low or h is high (strategic myopia dominates). Figure B1 illustrates these patterns: Panels (A)–(C) vary q while holding h fixed, and Panels (D)–(F) vary h while keeping q fixed.

[Figure B1 here]

These patterns in trading aggressiveness directly affect price efficiency. In the model with partially myopic quant, the equilibrium price efficiency is given by

$$\text{PE}_{\text{partially myopic quant}} = \frac{2\gamma h + 6}{(3\gamma h - 3q\gamma - q\gamma^2 h + 8)\Sigma_0}. \quad (\text{B-9})$$

Proposition B3 summarises how price efficiency responds to increases in quantitative investing when one firm is fully discretionary and the other has a partially myopic quant department.

Proposition B3. *When the quant department of firm B ignores its price impact with probability h , the effect of an increase in quantitative investing level γ on price efficiency depends on the ratio of signal quality q to the degree of strategic myopia h :*

- (i) *When $0 < q/h \leq a_1$, price efficiency decreases with the level of quantitative investing γ .*
- (ii) *When $a_1 < q/h < a_2$, price efficiency decreases with the level of quantitative investing γ if and only if $\gamma < \bar{\gamma}^*$.*
- (iii) *When $q/h \geq a_2$, price efficiency increases with the level of quantitative investing γ .*

When the ratio q/h is low ($0 < q/h \leq a_1$), indicating either low signal quality or high strategic myopia, price efficiency decreases with increased quantitative investing (see Figure B2). This is because the strategic myopia effect dominates when q is low or h is high (see Panels A and F of Figure B1). When q/h is high ($q/h \geq a_2$), suggesting either high signal quality or low strategic myopia, price efficiency increases with quantitative investing. In this case, the processing capacity effect dominates (Panels C and D of Figure B1). For moderate values of q/h ($a_1 < q/h < a_2$), price efficiency is non-monotonic in γ . It initially decreases when $\gamma < \bar{\gamma}^*$ but increases γ exceeds $\bar{\gamma}^*$, as in the baseline model.

[Figure B2 here]

Figure B1**Trading aggressiveness as a function of quantitative investing with partially myopic quant department of firm B (firm A is fully discretionary)**

This figure plots how trading aggressiveness measures respond to changes in the quantitative investing level γ when the quant department of firm B is partially myopic. Panels (A)–(C) vary signal quality q while holding $h = 0.5$ fixed, with $q = 0.04, 0.047$, and 0.057 respectively. Panels (D)–(F) vary h while keeping $q = 0.0611$ fixed, with $h = 0.32, 0.76$, and 0.96 . The blue dashed line represents the absolute value of the strategic myopia effect; the purple line represents the sum of the processing capacity and internalising effects. We set $\Sigma_0 = 1$ in all panels.

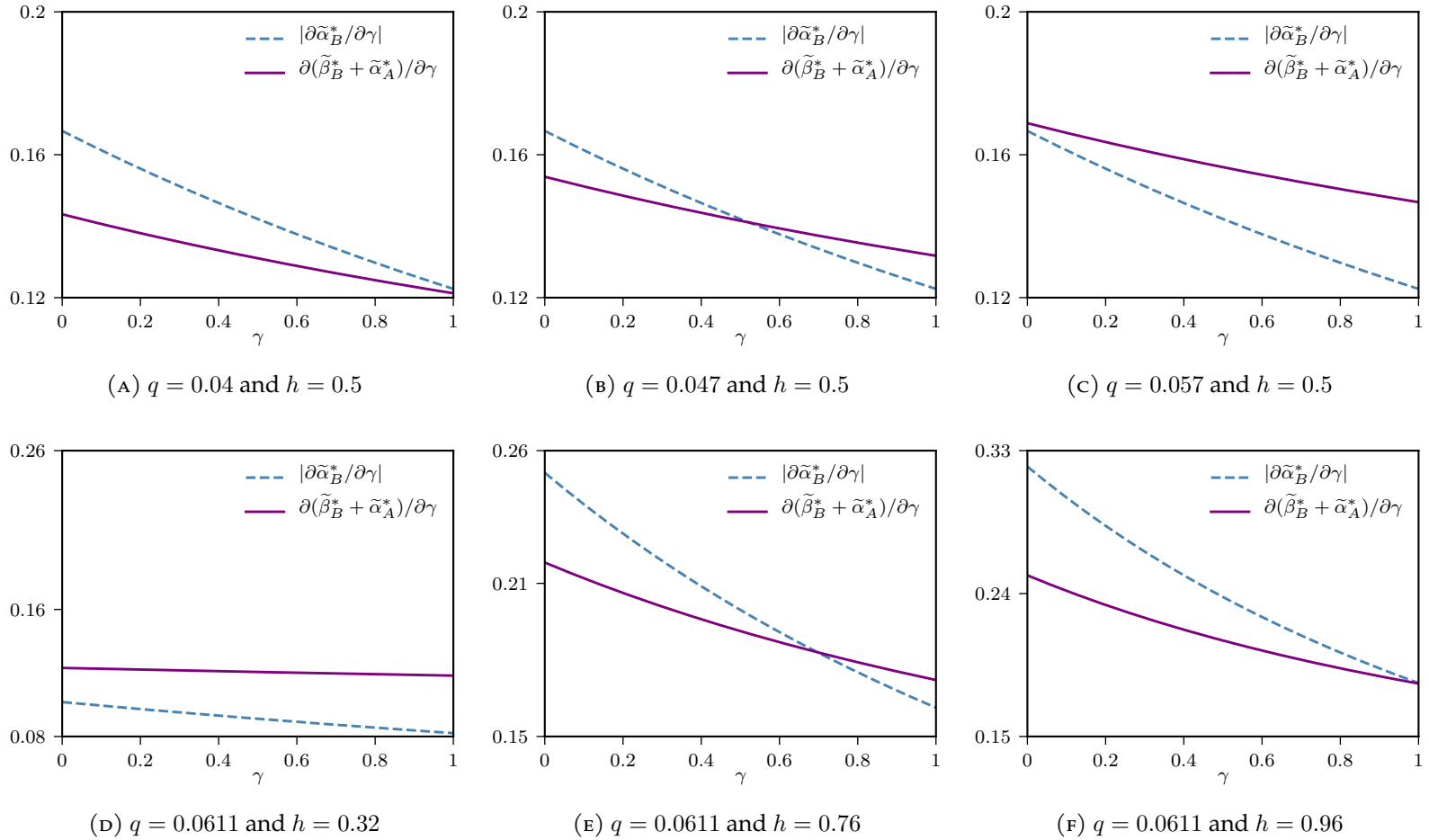


Figure B2**Price efficiency as a function of quantitative investing with partially myopic quant department**

This figure illustrates the impact of increasing the quantitative investing level γ on price efficiency when firm B 's quant department has partial strategic myopia. Panel (A) varies signal quality q while holding $h = 0.5$ fixed. The grey dashed line corresponds to $q = 0.04$, the blue line to $q = 0.047$, and the black dashed line to $q = 0.057$. Panel (B) varies the degree of strategic myopia h while holding $q = 0.0611$ fixed. The grey dashed line corresponds to $h = 0.32$, the blue line to $h = 0.76$, and the black dashed line to $h = 0.96$. The other parameter value is $\Sigma_0 = 1$.

